

Tailoring passivators for highly efficient and stable perovskite solar cells

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Abstract

There is an ongoing global effort to advance emerging perovskite solar cells (PSCs), and many of these endeavours are focused on developing new compositions, processing methods and passivation strategies. In particular, the use of passivators to reduce the defects in perovskite materials has been demonstrated to be an effective approach for enhancing the photovoltaic performance and long-term stability of PSCs. Organic passivators have received increasing attention since the late 2010s as their structures and properties can readily be modified. First, this Review discusses the main types of defect in perovskite materials and reviews their properties. We examine the deleterious impact of defects on device efficiency and stability and highlight how defects facilitate extrinsic degradation pathways. Second, the proven use of different passivator designs to mitigate these negative effects is discussed, and possible defect passivation mechanisms are presented. Finally, we propose four specific directions for future research, which, in our opinion, will be crucial for unlocking the full potential of PSCs using the concept of defect passivation.

Sections

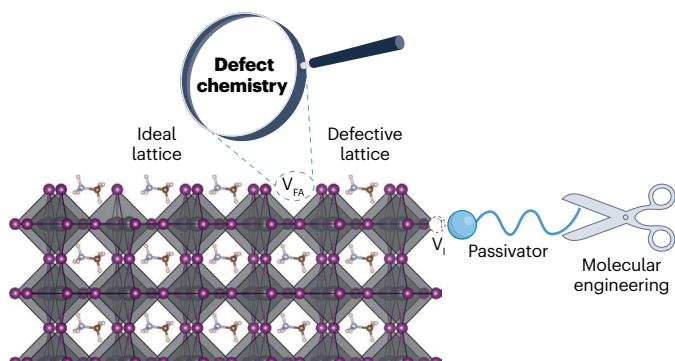
Introduction

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Introduction

Since their inception in 2009 (ref. 1), the power conversion efficiency (PCE) of perovskite solar cells (PSCs) has improved from 3.8% to a current certified record of 25.8% according to the National Renewable Energy Laboratory (Fig. 1a), with a lot more room for improvement given a Shockley–Queisser limit of ~33% efficiency for a perovskite with an ideal bandgap (E_g) of 1.34 eV (ref. 2).

The photoactive layer in PSCs consists of a metal halide perovskite (MHP) with the general formula ABX_3 , in which A represents a monovalent organic (such as methylammonium (MA^+) or formamidinium (FA^+)) or inorganic (mostly caesium (Cs^+)) cation, B is commonly Pb^{2+} or Sn^{2+} and X is a halide (mostly I^- and Br^- ; less-frequent Cl^-). In the photoactive phase of a perovskite, these components adopt a cubic crystal structure of corner-sharing BX_6 octahedra with A cations residing in the cavities surrounded by 12 halides forming a cuboctahedral coordination sphere (Fig. 1b). Compared with amorphous silicon (Si, barriers for nucleation and crystallization are 470 kJ mol^{-1} and 280 kJ mol^{-1} , respectively)³, the crystallization activation energy barriers of perovskites are considerably lower ($56.6\text{--}97.3\text{ kJ mol}^{-1}$)⁴, depending on the Pb precursors. Consequently, highly crystalline perovskite thin films with low defect densities can be prepared using different low-temperature solution-based techniques, considerably lowering production costs compared with Si.

An additional advantage is the possibility of incorporating mixtures of ions at the A, B and X sites. This allows for tuning the bandgap between 1.2 eV and 3.0 eV by compositional engineering⁵ (Fig. 1c). Specifically, replacing Pb^{2+} with Sn^{2+} generally leads to a narrowing of the bandgap⁶, whereas the E_g consistently increases in the order of $I^- < Br^- < Cl^-$ (refs. 7,8). Mixing A-site cations has a considerably smaller influence on the bandgap as the comparison between $MAPbI_3$ ($E_g = 1.5\text{--}1.6\text{ eV}$)⁹ and $FAPbI_3$ ($E_g = 1.48\text{ eV}$)¹⁰ shows, but it has been used extensively to improve the stability of perovskites¹¹, for example, in the so-called triple-cation ($Cs\text{-}MA\text{-}FA$) compositions¹².

The reason for the different effects of replacing or mixing B^- and X^- versus A-site ions can be understood from the electronic structure of MHPs. Computational studies on cubic $MAPbI_3$ have shown that the valence band maximum (VBM) is antibonding possessing approximately 3:1 $5p$ and $Pb\ 6s$ character, whereas its conduction band minimum (CBM) is dominated by weakly interacting (antibonding) $Pb\ 6p$ and $I\ 5p$ orbitals¹³ (Fig. 1d). These results have been qualitatively confirmed for other compositions including $CsPbX_3$ ($X=I, Br$ and Cl)¹⁴. Furthermore, relativistic effects lower the CBM explaining the lower bandgaps of Sn-based compared with Pb-based perovskites¹⁵. However, it is important to note that most other semiconductors have band edges with bonding (VBM) and antibonding (CBM) character making them vulnerable to the formation of deep defect states owing to interstitials and vacancies¹⁶. For A-site cations, their influence on the stability of a perovskite can be explained by considering their different sizes. This can be expressed with the Goldschmidt tolerance factor ($t = (r_A + r_X) / (\sqrt{2}(r_B + r_X))$), putting into relation the three ionic radii (r_A, r_B and r_X) of a perovskite¹⁷, with values for t between 0.9 and 1.0 generally indicating a stable cubic phase¹⁸.

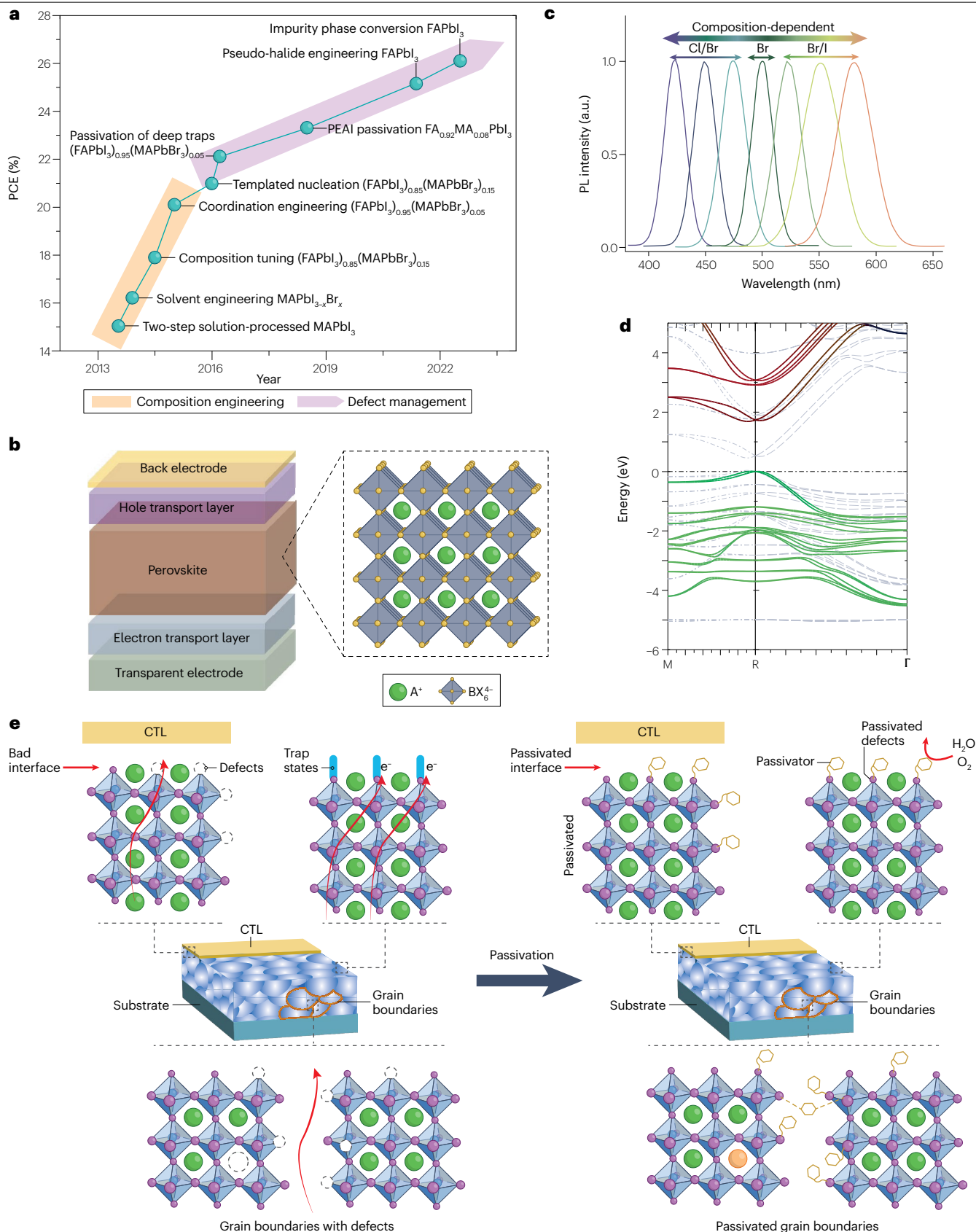
Importantly, MHPs possess a higher absorption coefficient ($\sim 10^5\text{ cm}^{-1}$) than most semiconductors such as Si ($10^3\text{--}10^4\text{ cm}^{-1}$)¹⁹, which is due to its direct bandgap and $Pb\ 6p\text{--}I\ 5p$ transitions in $MAPbI_3$ (refs. 13,20). Therefore, perovskites are highly efficient light absorbers, meaning that the photoactive layers of PSCs can be thinner (typically $300\text{--}600\text{ nm}$)²¹ compared with other photovoltaic (PV) technologies such as Si wafers ($200\text{--}500\text{ }\mu\text{m}$)²² contributing to a better performance. The low exciton binding energy of perovskites in the range of $2\text{--}75\text{ meV}$ ^{23,24} – depending on the measurement technique – is comparable to that of other inorganic semiconductors (Si: $\sim 14\text{ meV}$ ²⁵ and GaAs: $\sim 4\text{ meV}$ ²⁶) and allows for the thermal separation of photogenerated Frenkel excitons at ambient conditions, which further contributes to the high efficiency of PSCs by minimizing open-circuit voltage (V_{oc}) losses. Also, the low Urbach energy – the exponential increase of light absorbance with energy – of solution-processed polycrystalline perovskite thin films ($MAPbI_3$: $\sim 15\text{ meV}$)²⁷ is comparable to common single-crystalline inorganic semiconductors (Si: $\sim 11\text{ meV}$ ²⁷ and GaAs: $\sim 7.5\text{ meV}$ ²⁸), indicating a low density of deep traps. Furthermore, both electrons and holes have similar, long diffusion lengths ($>1\text{ }\mu\text{m}$)^{29–31} – greater than the thickness of a typical thin film in a PSC – owing to their similarly low effective mass³², which is an effect of the strong $s\text{--}p$ antibonding coupling mentioned before³³. This allows for efficient charge collection, which is important for achieving high PCEs. These long diffusion lengths are related to the high charge carrier mobility ($\sim 1\text{--}100\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$, depending on the measurement technique)^{34,35} in perovskites and remarkably low bi-molecular recombination rates, defying the Langevin limit³⁶.

To date, the record PCE for PSCs is still lower than the Shockley–Queisser limit – mainly owing to V_{oc} losses – but the extraordinary efficiencies achieved to date are made possible owing to the properties discussed so far. Since 2020 and apart from increasing PCE, more attention has also been paid to improving the stability of perovskite thin films and PSCs, which is a dominant obstacle for commercialization^{37–39}. A central approach for addressing both challenges is to reduce the number of crystallographic defects present in perovskite thin films (Fig. 1a), which have been linked to reducing PCE. The defects can produce trap states³⁹ – which are energy states sitting within the bandgap that trap charge carriers – that promote degradation, for example, by reaction with ambient moisture⁴⁰ and oxygen⁴¹, and facilitate ion migration⁴². In mixed-halide perovskites, defects have also been linked to the commonly observed photostimulated halide phase segregation, which rapidly reduces PCE under operating conditions^{43,44}.

Different strategies for reducing the number of defects such as optimizing the perovskite precursor solution or the fabrication process have been explored⁴⁵. One of the most prominent methods is the use of additives, the so-called passivators, which can either be added to the precursor or the antisolvent, but can also be applied in a subsequent step onto the as-prepared perovskite films^{37,39}. In addition to reducing the number of defects and manipulating the energy levels of remaining defects, passivators can also form physical barriers against moisture and oxygen uptake, while reducing the loss of volatile perovskite

Fig. 1 | Introduction to metal halide perovskites and the concept of passivation. **a**, Improvement in perovskite solar cell efficiency over time and important technological advances. **b**, Stack of an $n\text{--}i\text{--}p$ -type perovskite solar cell with a zoom into the structure of an ideal cubic metal halide perovskite (MHP). **c**, Bandgap tuning in Cs-based inorganic MHPs as a result of varying the halide⁸. **d**, Calculated band structure of $MAPbI_3$ obtained using the quasiparticle

self-consistent GW formalism¹³. The valence band minimum was set at 0 eV. The contributions from $I\ 5p$ (green) and $Pb\ 6p$ (red) orbitals are shown. **e**, Concept of passivation of the surfaces, interfaces and grain boundaries of MHPs. CTL, carrier transport layer; PCE, power conversion efficiency; PEAI, phenylethylammonium iodide; PL, photoluminescence. Part c adapted with permission from ref. 8, Wiley. Part d adapted with permission from ref. 13, ACS.



components, preventing charge-transport-induced degradation and improving the perovskite-charge-transport layer interfaces (Fig. 1e). It has furthermore been established that defect formation during device operation can be reversible in the dark^{46,47}.

In this Review, we discuss the defect chemistry of MHPs, including the density and nature of defects as well as their effects regarding efficiency and stability. On the basis of this, we provide an overview of different strategies for defect passivation including neutralization of charged defects via formation of coordination bonds, passivation through supramolecular interactions, redox chemistry control and phase conversion. We present different classes of passivators from small organic molecules to polymers and organic ammonium halides and deduce general design principles to aid the development of improved next-generation passivators.

Defect chemistry in halide perovskites

Defects have a central role in the properties of semiconductors. The optoelectronic properties of classical inorganic semiconducting materials such as crystalline silicon, GaAs, CdTe and CuInGaSe (CIGS) are highly sensitive to defects, and great care must be taken to keep the density of defects low to achieve the desired performances. Single crystals of MHPs have been reported to have very low effective defect densities ($\leq 10^{12} \text{ cm}^{-3}$)⁴⁸ compared with inorganic semiconductors (polycrystalline Si⁴⁹: 10^{13} – 10^{14} cm^{-3} ; GaAs⁵⁰: 10^{13} – 10^{15} cm^{-3} ; CdTe⁵¹: 10^{13} – 10^{16} cm^{-3} and CIGS⁵²: 10^{11} – 10^{15} cm^{-3}), and polycrystalline thin films suggested to be even greater (10^{15} – 10^{16} cm^{-3})⁴⁸. It has, however, been argued that thin-film values should be considered with caution as they are likely to represent the upper limits owing to the low sensitivity of the used experimental techniques⁴⁸. Although this question is still debated, it is commonly accepted that despite the perceived defect tolerance and self-healing capabilities of MHPs, appropriate defect management is essential for achieving the efficiencies and stabilities required for commercial exploitation of PSCs.

There are two kinds of defects in MHPs – point and extended defects. Point defects can be further divided into two categories of those only involving components of the perovskite material (native point defects) and those involving foreign species (impurities) (Fig. 2a). These crystallographic imperfections lead to the formation of localized electronic states whose energy levels are different from the carrier transport bands and can, therefore, act as traps for photogenerated free carriers, severely impacting the efficiency of PSCs.

Point defects

According to the laws of thermodynamics, native defects will always be present in a material when $T > 0 \text{ K}$, reducing the free energy of the system as the entropy increases. Yin et al.⁵³, however, demonstrated that the dominating type of defects in MHP structures depends on the growth conditions. As experiments cannot provide an atomistic picture of the defects present in MHPs, computational studies have been widely used as the preferred method for determining formation energies of point defects and the positions of the resulting electronic trap states relative to the bandgap⁵⁴. Most studies have been conducted on MAPbI₃, in which 12 native point defects have to be considered: three vacancies (V_{MA} , V_{Pb} and V_{I}), three interstitials (MA_i , Pb_i and I_i), two cation substitutions (MA_{Pb} and Pb_{MA}) and four antisite substitutions (MA_i , Pb_i , I_{MA} and I_{Pb})⁵³ (Fig. 2a). Yin et al. identified eight of them, V_{MA} , V_{Pb} , V_{I} , MA_i , I_i , MA_{Pb} , Pb_{MA} and MA_i , as having low energy of formation, in at least one of the studied growth conditions (I-rich and Pb-poor, neutral, Pb-rich and I-poor). The remaining four, Pb_i , Pb_i , I_{MA} and I_{Pb} , are thermodynamically

unfavoured⁵³ (Fig. 2b). The dominant *p*-type and *n*-type defects are V_{Pb} and MA_i (Table 1), respectively, and good intrinsic *p*-type and *n*-type conductivity can be obtained by varying the growth conditions to favour the formation of one over the other. Conveniently, the easily formed and thus dominant defects exclusively result in electronic states residing within the band structure or shallow charge carrier traps within $-k_{\text{B}}T$ of the VBM or CBM, whereas trap states localized deep inside the bandgap are formed from less thermodynamically stable defects (Fig. 2c). Shallow trap states slow down the movement of free charge carriers as, once trapped, they can be thermally re-excited back into the band structure (Fig. 2d). According to this model, no considerable loss of initial PCE is observed for shallow traps, but they can limit the maximum current density produced by the device. This is often described by the multiple-trapping-and-release model⁵⁵. The tolerance of MAPbI₃ to V_{MA} and V_{I} defects has been verified experimentally⁵⁶. Conversely, deep electronic traps act as Shockley–Read–Hall⁵⁷ centres for non-radiative recombination leading to considerable efficiency losses mainly by reducing the V_{OC} of devices⁵⁸ (Fig. 2d). $V_{\text{H}}(\text{N})$ – a hydrogen vacancy created by removing H from an NH₃ group of MA – has been identified as a strong non-radiative recombination centre with low formation energy in MAPbI₃ (ref. 59).

The I_i defect is also of considerable importance for the performance of MAPbI₃, owing to its redox chemistry, which can be neutral or charged – either negatively or positively. Although I_i is among the most stable point defects in MAPbI₃ under neutral conditions (1:1 ratio of lead iodide (PbI₂) and MAI), it is easily oxidized to the corresponding I_i^+ defect at Fermi energies close to VBM. I_i^- and I_i^+ defects can trap photogenerated holes and electrons, respectively, to form the neutral I_i^0 defect^{60,61}. The crystal lattice rearrangement associated with this process is small in the case of I_i^- , whereas a considerable lattice relaxation is found for I_i^+ . I_i^0 formed through the hole capture of I_i^- is thus easily reduced back to I_i^- , whereas I_i^0 formed through the electron capture of I_i^+ constitutes a long-lived filled trap state⁶⁰. This observation has been suggested to be linked to the extremely long lifetimes (tens of microseconds⁶²) of filled electron traps in MAPbI₃. Exposure of MAPbI₃ to I₂ vapour leads to the formation of pairs of I_i^- and I_i^+ defects⁶³. In a reversal of this process, formation of I_i^0 by hole or electron capture at I_i^- or I_i^+ can lead to the subsequent formation of I_2 , which can escape spontaneously from MAPbI₃.

The defect chemistry of MAPbBr₃ is similar to that of MAPbI₃ with MA_i and negatively charged Br_i having been identified as the most stable defects⁶¹. However, the V_{Pb} defect is less stable in MAPbBr₃, and Br_i^+ has a higher defect formation energy compared with I_i^+ in MAPbI₃ owing to the lower electronegativity of iodine compared with bromine.

The dominance of shallow defects in MAPbI₃ is due to the electronic structure at VBM and CBM⁵³ (Fig. 1d). Antibonding Pb 6s to I 5p interactions leads to an increased VB bandwidth and a raised VBM resulting in most *p*-type defects being located inside or close to the VB. Weak Pb 6p–I 5p coupling at CBM results in a predominantly ionic character, in which energy states are only weakly affected by defects. Furthermore, the strong relativistic effect of Pb lowers the CBM of Pb perovskites¹⁵.

Similar results have also been found for FAPbI₃, which has replaced MAPbI₃ as the material of choice for obtaining record PCEs owing to its lower bandgap, which is closer to the Shockley–Queisser optimum. Antisite defects FA_i and I_{FA} create deep-level trap states, but they have substantially lower formation energies, making FAPbI₃ a less defect-tolerant material⁶⁴. By contrast, hydrogen vacancies $V_{\text{H}}(\text{C})$ (H removed from the methyl group in FA) have a higher formation

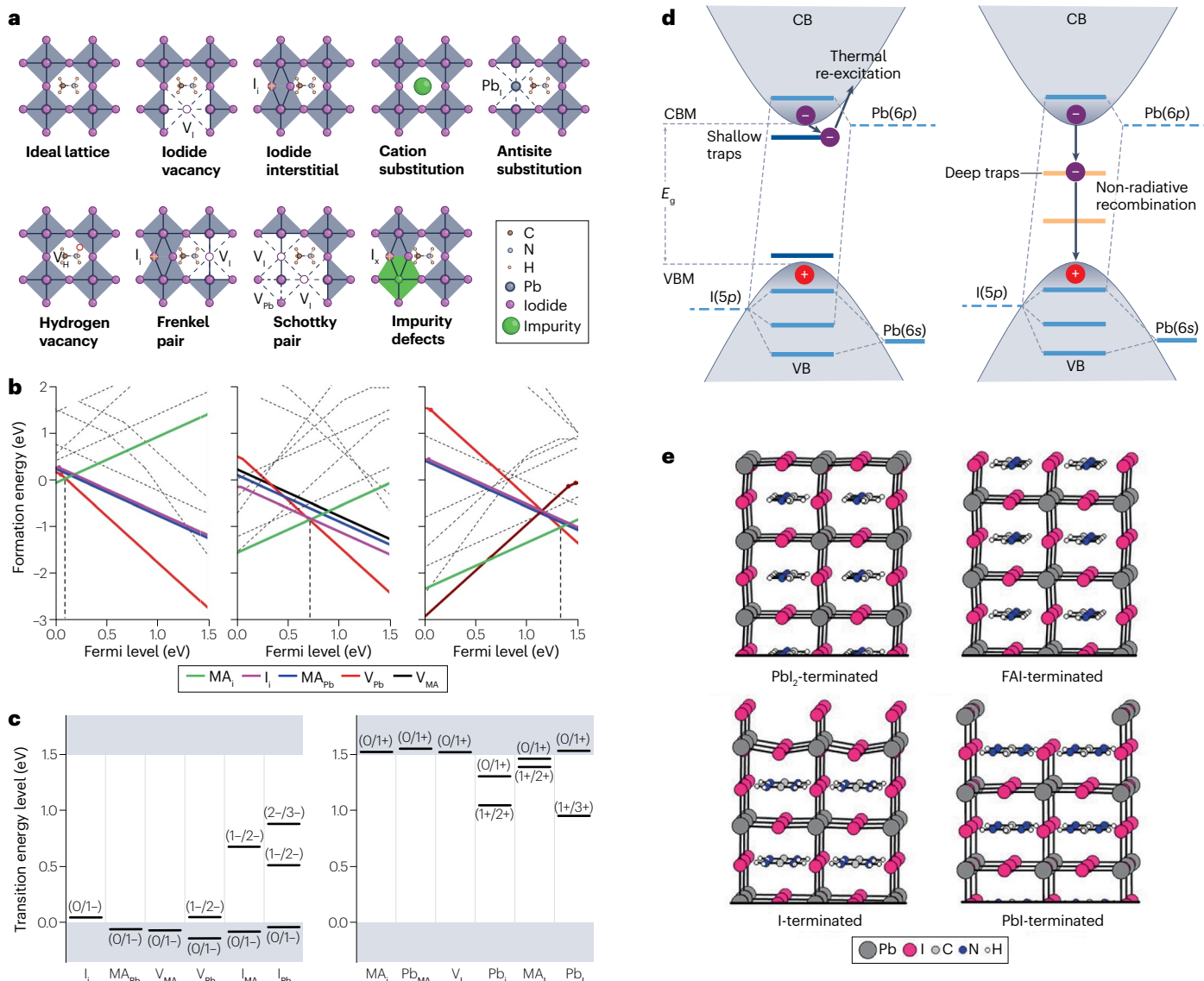


Fig. 2 | Point and extended defects in metal halide perovskites. a, Point defects and combinations of point defects in an MAPbI₃ lattice. **b**, Formation energies of defects under three different conditions in MAPbI₃: I-rich and Pb-poor (left), moderate (middle) and I-poor and Pb-rich (right)⁵³. **c**, Transition energy levels of intrinsic acceptors (left) and intrinsic donors (right) in MAPbI₃ (ref. 53). **d**, Scheme illustrating deep and shallow charge carrier traps. Note that trapping followed by thermal

re-excitation (shallow traps) and non-radiative recombination (deep traps) affects holes in an analogous way to electrons shown here. **e**, Schematic representation of the PbI_2 -terminated, FAI-terminated, I-terminated and PbI -terminated (001) surfaces of FAPbI₃⁹⁴. CB, conduction band; CBM, conduction band minimum; VB, valence band; VBM, valence band maximum. Parts **b** and **c** were adapted with permission from ref. 53, AIP. Part **e** was adapted with permission from ref. 93, ACS.

energy and smaller capture cross-section than the related hydrogen vacancies in MAPbI₃ (ref. 59).

Dunfield et al.⁶⁵ have suggested a correlation between defect tolerance and the tolerance factor of perovskites. Although MAPbI₃ (ref. 53), CsPbBr₃ (ref. 66), CsSnI₃ (ref. 67), CsSnBr₃ (ref. 18), CsSnCl₃ (ref. 18) and MAPbBr₃ (ref. 68) are defect-tolerant with factors roughly between 0.82 and 0.85, FAPbI₃ (ref. 64), CsPbI₃ (refs. 69,70) and FASnI₃ (ref. 70) lie outside this range and have been suggested to be less tolerant. Replacing approximately half of the FA⁺ cations in FAPbI₃ with MA⁺ to achieve an appropriate tolerance factor increases the formation energies of detrimental FA_i and I_{FA} defects⁶⁴. In the case of CsPbI₃, the energetically

favourable formation of deep trap states related to Pb_i and I_{Pb} antisite defects is compensated by their small capture cross-sections owing to strong anharmonicity⁷¹. Formation of deeper trap states related to the intrinsic defects in δ -CsPbI₃ compared with those of γ -CsPbI₃ have furthermore been linked to a weaker antibonding character in the VBM of the former⁶⁹.

Native point defects can also occur in pairs of defects with opposite charges, such as in Frenkel and Schottky defects (Fig. 2a). On the basis of theoretical^{72,73} and experimental results⁷⁴, iodine vacancies (V_I) and interstitials (I_I) seem to be the two most abundant defects in MAPbI₃ and are likely generated as Frenkel pairs. In FAPbI₃, the Frenkel pair of

Table 1 | The formation energies (eV) of 12 native point defects in MAPbI₃ at chemical potential points A, B and C shown in Fig. 2b (ref. 53)

	I _i	MA _{Pb}	V _{MA}	V _{Pb}	I _{MA}	I _{Pb}	MA _i	Pb _{MA}	V _i	Pb _i	MA _i	Pb _i
A	0.23	0.28	0.81	0.29	1.96	1.53	1.39	2.93	1.87	4.24	3.31	5.54
B	0.83	1.15	1.28	1.62	3.01	3.45	0.93	2.51	1.27	2.91	2.25	3.62
C	1.42	1.47	2.01	2.68	4.34	5.10	0.20	1.74	0.67	1.85	0.93	1.97

V_{Pb} and Pb_i has been calculated to form a stable and deep trap, even though both defects individually are unstable⁷⁵. The same study also showed this effect for the pair FA_{Pb} and Pb_{FA}, and similar observations have also been reported for the FA_{Pb} and FA_i, I_i and Pb_i, V_{Pb} and FA_i and V_{Pb} and Pb_i defect pairs in FAPbI₃ (ref. 75). Schottky defects, consisting of pairs of anion and cation vacancies, have also been demonstrated to have a key role in MAPbI₃ (ref. 74) and FAPbI₃ (ref. 76).

Impurities – especially metal cations introduced into the perovskite layer during preparation or operation via its migration from Li-doped spiro-OMeTAD or a metal electrode – can also introduce deep-level traps in the form of substitutional or interstitial defects (Fig. 2a) acting as centres for non-radiative recombination. A density functional theory study by Ming et al.⁷⁷ showed that a range of metals, including Au, Co, Ni and Cr, have a low formation energy for deep-level defects in MAPbI₃. Migration of Au from the back electrode through the hole transport material (HTM) into the perovskite layer leading to irreversible device degradation has been demonstrated experimentally⁷⁸. Efficiency losses have also been shown for a MAPbI₃-based device following incorporation of Fe impurities into the perovskite layer⁷⁹. However, the tolerance to this impurity was reported to be substantially higher compared with *p*-type silicon devices⁸⁰. High sensitivity to Fe²⁺ has also been reported in a different study⁸¹. In addition, the authors there found that up to 6% of Pb in MAPbI₃ could be replaced with Co²⁺ improving efficiency compared with the reference device, in contradiction to the earlier report of Ming et al.⁷⁷. On the contrary, Bi³⁺ has been found to lower V_{OC} of devices even when present in parts per million (ppm) quantities⁸².

Extended defects

Compared with point defects, extended defects are of a higher dimensional order, similar to the surface of a crystal or the boundary with a neighbouring grain. These parts of a crystal are naturally defective as they represent a discontinuity of the crystal lattice leading to undercoordination of the exposed ions. Experimental characterization of the atomistic structures of surfaces and grain boundaries (GBs) of MHP thin films – relevant to defects and the related local band structure – remains sparse. However, their effects on MHP film properties have been observed. A tenfold difference in charge carrier lifetimes between the surface (4.9 ns) and bulk (109 ns) of MAPbI₃ single crystals⁸³, as well as an increase of the trap density by more than two orders of magnitude between the bulk and surface regions for single-crystal and polycrystalline MHP thin films, has been reported⁸⁴. Because of this critical influence of the surfaces and GBs on film properties, passivation of these regions is critical for obtaining high-efficiency devices.

Thermodynamic stability defines which facets of the crystal structure are dominant on the surface of MHP films. Density functional theory calculations have shown that the (110) and (001) surfaces of tetragonal MAPbI₃ (ref. 85) and the (001) surface of cubic FAPbI₃ (ref. 86) are the most stable, which in the case of the former has also been confirmed experimentally^{87,88}. This observation has been linked to their

nonpolar character^{87,89}. These facets can furthermore have different terminations (such as alkylammonium iodide or PbI₂) governing which crystal component will be exposed at the surface, as well as the nanoscale structure of the surface (Fig. 2e). For MAPbI₃, initial studies focused on PbI₂-rich surface terminations owing to the volatility of MAI⁸⁵, which was expected to be lost during annealing. Later experimental investigations using scanning tunnelling microscopy found MAX-terminated surfaces for MAPbI₃ (X=I)⁹⁰ and MAPbBr₃ (X=Br)⁹¹. MAI termination of solution-processed MAPbI₃ was also the conclusion of a combined ultraviolet photoelectron spectroscopy and meta-stable-atom electron spectroscopy study on tetragonal solution-processed MAPbI₃ films⁹². A computational study on FAPbI₃ has suggested the corresponding FAI-terminated surface to be thermodynamically preferred. However, conversion to an I-terminated surface has also been considered following loss of volatile FA⁹³.

Owing to the soft lattice of MHPs⁹⁴, however, the precise structure of their surfaces is generally accepted to be considerably more complicated than simply a cut through the 3D lattice at a certain position and along a certain orientation. For example, weak hydrogen bonds of the organic cations⁹⁵ make rearrangements of the exposed A-site cations possible⁹⁰. Furthermore, a lower formation energy of surface defects in MAPbI₃ (ref. 61) and FAPbI₃ (ref. 93) compared with the bulk leads to a more heterogeneous composition and structure. According to calculations, the formation energies of point defects and the depths of the associated trap states are highly dependent on the termination of MHPs, with MAI-terminated⁹⁶ and FAI-terminated surfaces⁹³ being the most defect-tolerant in MAPbI₃ and FAPbI₃, respectively. Comparing the formation energies of bulk and surface defects, only the formation of V_{MA} was found to be favourable on the MAI-terminated surface of MAPbI₃, resulting in a shallow trap. By contrast, V_{Pb}, V_i and I_i are more stable on the PbI₂-terminated surface⁹⁶. On the FAI-terminated surface of FAPbI₃, V_i is stable for a *p*-type material, whereas V_{FA} and I_i defects become dominant in an *n*-type host material. For the PbI₂-terminated surface, FA_i, Pb_i and FA_i defects have the lowest formation energies in *p*-type FAPbI₃, with FA_{Pb} and I_i becoming more stable under *n*-type conditions⁹³. We note, however, that formation energies are dependent on the growth conditions⁹⁷ and Fermi levels⁹³. In addition to point defects, larger defects such as Pb and I clusters have been observed on surfaces^{98–100}.

At GBs, neighbouring perovskite grains – whose lattices are misaligned relative to each other – meet and form an interface featuring many unsaturated ions and dangling bonds. Although Kelvin probe force microscopy and conductive atomic force microscopy measurements have shown that charges can be efficiently separated at GBs in MAPbI₃ (ref. 101), confocal fluorescence microscopy measurements correlated with scanning electron microscopy imaging revealed a lower photoluminescence (PL) intensity at GBs, compared with the centre of the grains¹⁰². This effect was attributed to more efficient non-radiative recombinations. In a separate study, an increased grain size in MAPbI₃ and Cs_{0.1}FA_{0.9}PbI_{2.9}Br_{0.1} perovskites was shown to

correlate with an increased V_{OC} , confirming the detrimental impact of GBs on device performance¹⁰³. Yin et al.¹⁰⁴ found that GBs in MAPbI₃ could increase the effective hole mass by introducing defect levels around the VBM. For CsPbI₃, iodine interstitials in both their positively (I_i^+) and negatively (I_i^-) charged state are more stable at GBs compared with the bulk, resulting in a driving force for these mobile defects to accumulate at GBs¹⁰⁴. As both these defects act as recombination centres in CsPbI₃, GBs contribute to PCE losses in this material. Clustering of traps at GBs has also been observed experimentally in a triple-cation mixed-halide perovskite¹⁰⁵. As a result, the defect density at surfaces and GBs is orders of magnitude higher compared with the bulk as seen in the comparison between MHP thin films (10^{15} – 10^{16} cm⁻³) and single crystals ($\leq 10^{12}$ cm⁻³)⁴⁸.

As we discussed previously for bulk defects, surface defects introduce localized trap states, which can act as centres for non-radiative charge carrier recombination, representing a major loss mechanism in PSCs. Electron–phonon coupling at surfaces and interfaces has also been put forward as the origin of below bandgap exciton and charge traps¹⁰⁶. In this interpretation, the entire surface of a 3D perovskite film can be regarded as trapping the charge or exciton. Charge carrier trapping at interfaces between photoactive perovskites and charge-transport layers is deemed to lead to faster recombination than in the bulk¹⁰⁷. GBs of MAPbI₃ films have also been shown to have a higher work function compared with bulk grains, indicating a potential barrier for electron transport between grains¹⁰⁸. Higher bandgaps of the surface regions compared with the bulk have been observed in MAPbI₃ and MAPbBr₃ single crystals, possibly caused by strong lattice deformation⁸³.

In conclusion, surfaces forming interfaces with neighbouring charge-transport layers and GBs are responsible for the majority of defect-promoted performance losses in PSCs. Their passivation is, therefore, essential for obtaining PSCs with the efficiencies and stabilities necessary for commercialization.

Influence of defects on extrinsic perovskite stability

The stability of PSCs is severely impacted by a range of external factors such as illumination, temperature, moisture and oxygen. Many of the associated degradation mechanisms are linked to – or even dependent on – the presence of the defects discussed in the previous section. For example, defect-promoted ion migration is substantially accelerated under illumination¹⁰⁹, leading to effects such as hysteresis and halide phase segregation (Box 1). Charges trapped at defective surfaces and defects provide infiltration pathways for water into the perovskite film triggering its irreversible degradation¹¹⁰. With regard to oxygen, Meggiolaro et al. demonstrated that molecular oxygen reversibly adsorbs to and diffuses into an MAPbI₃ film through iodine vacancies⁴² – the formation of which is promoted under illumination¹¹¹. The presence of O₂ in a perovskite crystal leads to the formation of new energy states below the conduction band⁴², which can trap photoelectrons leading to the formation of highly reactive superoxides (O₂⁻)^{42,112}. This species can act as an acceptor dopant⁴² and deprotonate MA⁺, triggering the decomposition of MAPbI₃ (ref. 112). In contrast to MAPbI₃, MAPbBr₃ benefits from an improved stability towards O₂⁻, probably as a result of its higher thermodynamic stability¹¹³.

Defect passivation mechanisms

With respect to the different types of defect in MHPs, different passivation mechanisms have been investigated (Table 2). Understanding them is critical for the rational design of better performing passivators.

Neutralization of charged defects

The ionic nature of MHPs imposes different requirements for defect passivation compared with covalently bonded semiconductors such as Si, as the defects in MHPs can be charged, either positively or negatively. Therefore, ions with charges complementary to a given defect can bond with the defect and effectively annihilate the defect-induced charge traps (Fig. 3a). Through ionic bonding and other electrostatic interactions, metal ions and organic cations have proven to be effective passivators for the negatively charged defects of MHP materials, such as undercoordinated I⁻, antisite Pbl₃⁻ and FA⁺ vacancies³⁹. The potential passivating effect of metal ions was first reported when Bi et al. found that sodium ions, Na⁺, passivate negatively charged defects, either by adsorption in MA⁺ vacancies at the GBs – owing to their similar size and isovalent charge relative to MA⁺ – or by forming an ionic bond with undercoordinated halides¹¹⁴. Beyond metal ions, research has also been conducted on organic cationic passivators, such as those featuring ammonium (R–NH₃⁺) functionalities. Unlike amines (R–NH₂), which coordinate to positively charged defects through the lone pair

Box 1

Introduction of ion migration in metal halide perovskites

Ion migration is equivalent to the movement of charged defects under the influence of an external bias and is, therefore, directly linked to the presence of crystalline defects – especially vacancies and interstitials promoting halide hopping²⁶¹. Under the influence of an electric field, positively (such as V_i and MA_i) and negatively charged defects (such as V_{MA} and I_i) migrate to hole transport layer and electron transport layer interfaces, respectively, causing an electric field screening effect and potentially hampering charge extraction^{72,261–263}. Surfaces and grain boundaries – regions with elevated defect concentrations – have been shown to be the primary channels for ion migration^{42,264}. Defect migration activation barriers spanning a huge range from ~0.1 eV to ~0.6 eV have been obtained computationally for the migration of iodide defects in MAPbI₃ (refs. 42,72,261,265–268), broadly matching experimental results^{261,265,267}. According to calculations, halide defects migrate faster than A-site cations, which themselves are faster than Pb defects^{72,261,269}. Illumination lowers the activation barriers for defect migration^{109,270} and has also been suggested to introduce additional iodide vacancies, following generation of elemental iodine after iodide oxidation by photogenerated holes¹¹¹. Subsequent loss of I₂ can lead to permanent material degradation. Also, interstitial hydrogen²⁷¹ and impurities such as Na⁺ and Li⁺ have been shown to migrate through metal halide perovskite films²⁷². Current density–voltage hysteresis has often been linked to ion migration^{72,273–277}, which has also been invoked to explain phase segregation in mixed-halide perovskites under illumination^{44,278–281}. Reactions of the migrating iodide with oxidized spiro-OMeTAD^{282–284} or metal electrodes^{285,286}, as well as band bending²⁵⁷ upon redistribution of charge defects, are further concerns regarding the performance of perovskite solar cells related to ion migration.

Table 2 | Systematic comparison of different types of passivator

Passivation mechanism	Specific mode of operation	Passivated defects	Examples of passivators
Neutralization of charged defects	Formation of ionic bonds or other electrostatic interactions	Negatively charged defects (undercoordinated I ⁻ , antisite, FA ⁺ vacancies)	Metal ions (Na ⁺ , K ⁺ , Rb ⁺ , Cd ²⁺); organic ammonium cations (R-NH ₃ ⁺)
		Positively charged defects (undercoordinated Pb ²⁺ , halide vacancies (V _h))	Halides (F ⁻ or Cl ⁻); polyatomic anions (such as R-COO ⁻ , SCN ⁻ , BF ₄ ⁻)
Supramolecular interactions	Formation of halogen bonds	Undercoordinated halide anions	Iodopentafluorobenzene
	Formation of hydrogen bonds	Negatively charged halide vacancies (V _h)	2,5-Thiophenedicarboxylic acid; crown ethers (such as DB24C8)
Oxidation management	Suppression of halide oxidation to molecular X ₂	X ₂ (such as I ₂)	Eu ³⁺ and Eu ²⁺ ion pair; benzoquinone; benzylhydrazine hydrochloride
	Suppression of reduction of Pb ²⁺ to metallic lead (Pb ⁰)	Pb ⁰	Eu ³⁺ and Eu ²⁺ ion pair
	Suppression of oxidation of Sn ²⁺ to Sn ⁴⁺	Sn ⁴⁺	SnF ₂ ; Sn powder
Phase conversion	Conversion of impurity phases into photoactive 3D MHPs	Excess PbI ₂ ; non-perovskite phases (6H, 4H, 2H)	CsI-DB ₂₁ C ₇ complex
	Conversion of impurity phases into low-dimensional MHPs	Excess PbI ₂	RbCl; organic ammonium halides (such as BAI, PEAI, OAI)

BAI, butyl ammonium iodide; MHP, metal halide perovskite; OAI, *n*-octylammonium iodide; PEAI, phenylethylammonium iodide.

of the nitrogen, R-NH₃⁺ cations passivate negatively charged defects through electrostatic interactions including ionic bonding and hydrogen bonding³⁷. Despite considerable developments in the use of cationic passivators, anionic passivation alternatives, which could potentially interact with undercoordinated Pb²⁺ ions and halide vacancies, remain underexplored. To date, reported anions for defect passivation of perovskites include halides (F⁻ or Cl⁻)^{115,116} and small polyatomic anions (such as R-COO⁻, SCN⁻ and BF₄⁻)^{117,118}. Zwitterions possess both positively and negatively charged functional groups, a unique property that has proved beneficial in improving PV performance of PSCs^{119,120}. One of the most promising applications of zwitterions is the ability to simultaneously passivate negatively and positively charged ionic defects using both mechanisms discussed so far.

Supramolecular interactions

Undercoordinated halide anions create superficial defects on the surface of MHP grains, which can trap positive charges and holes, promoting non-radiative charge recombination. Halogen bonding can provide passivation of these undercoordinated halide anions and thus enhance the efficiency and stability of devices. The first example of MHP surface defect passivation through halogen bonding was reported by Abate et al.¹²¹. They used iodopentafluorobenzene (IPFB) to passivate undercoordinated halide ions (Fig. 3b), which led to a considerable improvement of performance in PSCs. IPFB can interact with I⁻ ions through halogen bonding, shielding their electrostatic charge and reducing the accumulated charge at the interface, thereby suppressing hole recombination and promoting charge transfer at the interface between the perovskite and the hole transport layer. The general applicability of this concept was further demonstrated by Ruiz-Preciado et al.¹²², who introduced 1,2,4,5-tetrafluoro-3,6-diiodobenzene as a bifunctional supramolecular modulator owing to its hydrophobicity and ability to bind strongly to MHPs through halogen bonding. When added to the precursor solution or used as an interlayer between MHPs and a charge-transport material, it resulted in an improved V_{OC} and enhanced

long-term operational stability. This work also provided fascinating insights into the atomic-level interaction between MHP and passivator, owing to solid-state NMR analyses and theoretical simulations.

Halide-vacancy defects in MHPs can also be passivated by forming hydrogen bonds between organic cations and supramolecular agents (such as crown ethers^{123–125}, acids¹²⁶ and cyclodextrins¹²⁷). A good example are crown ethers^{123,128}, which induce the formation of host–guest complexes on the surface of FA-rich MHP films after post-treatment, reducing the concentration of surface electronic defects, thus suppressing non-radiative recombination in PSCs. Incorporation of a polymeric adhesive with a high density of hydrogen-bonding functionalities has been reported to modulate the crystallization of MHP films, creating fewer trap states and an improved mechanical stability in highly efficient, flexible PSCs¹²⁹. The facile design of supramolecular passivators opens a new avenue to improve the efficiency and stability of MHP-based optoelectronic devices through halogen or hydrogen bonding.

Oxidation management

Owing to the soft nature of MHPs, chemical oxidation can occur in perovskite materials when exposed to various excitation stresses, which ultimately induces performance deterioration of PSCs as mentioned previously^{38,130}. First, halides such as I⁻ can easily be oxidized to I⁰, serving as carrier recombination centres as well as initiating chemical chain reactions accelerating the degradation of MHPs¹³¹. The oxidation of halide ions can be suppressed by adding reducing agents (such as benzoquinone¹³², piperidinium salts¹¹⁸ and benzylhydrazine hydrochloride¹³³) to the MHP precursor solution. Meanwhile, B-site metal ions can also undergo redox reactions upon exposure to harsh ageing conditions. Pb²⁺ is prone to be reduced to metallic Pb⁰ upon heating or illumination¹³⁴. Pb⁰ is the primary deep defect state that severely degrades the performance of PSCs, as well as their operational stability^{98,134}. Several attempts have been reported to suppress the formation of Pb⁰ or Sn⁴⁺ in a sustainable manner, such as optimizing film processing and additive engineering. A successful strategy reported by

Wang et al. is to use the europium ion pair Eu^{3+} – Eu^{2+} acting as a ‘redox shuttle’ selectively oxidizing Pb^0 and reducing I^0 defects simultaneously in a cyclical process¹³⁵ (Fig. 3c).

Sn-based MHPs have great potential as next-generation environment-friendly absorbers or narrow-bandgap absorbers for tandem applications, but their performance is hampered by undesirable Sn^{2+} to Sn^{4+} oxidation¹³⁶. Generally, Sn^{2+} is easily oxidized to Sn^{4+} upon exposure to oxygen¹³⁶. Studies have shown that Sn^{2+} can even be oxidized when these materials are processed in an inert atmosphere with low O_2 content

(<0.1 ppm)¹³⁷. Sn^{4+} is the primary deep-defect state and charge carrier recombination centre in these materials, leading to substantial losses of PV performance and operational stability of Sn-based PSCs^{136,138}. These results suggest the need for strategies to prevent such reactions and shed light on other solution instabilities beyond the formation of Sn^{4+} that must be mitigated to achieve high-performance Pb-free MHP optoelectronics. Innovative approaches, including the use of dopant additives^{139,140}, a reducing atmosphere¹⁴¹ and comproportionation reactions¹⁴², have been developed to effectively inhibit the oxidation of Sn^{2+} .

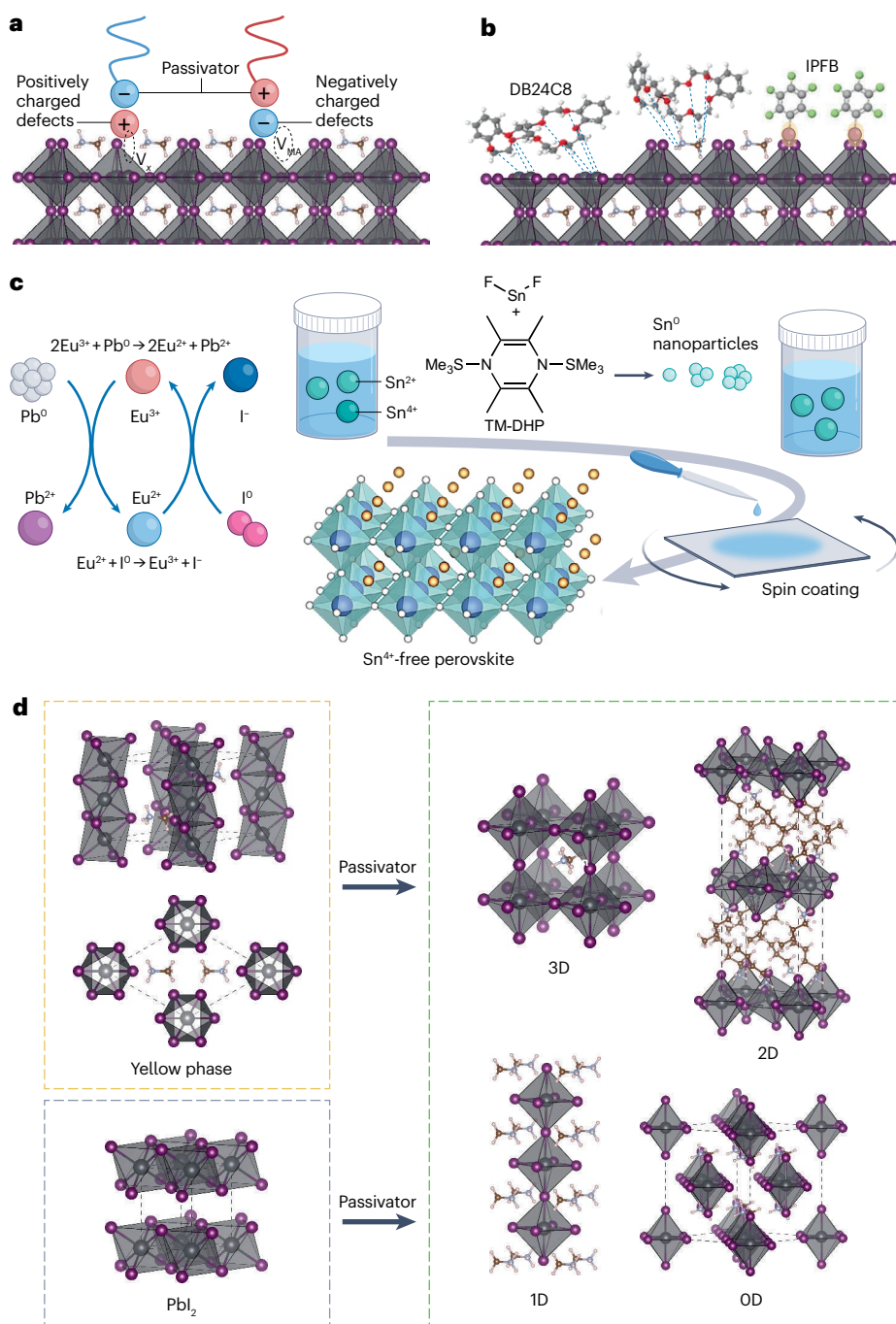


Fig. 3 | Defect passivation mechanisms in perovskite materials. **a**, Neutralization of charged defects. **b**, Supramolecular interactions. **c**, Oxidation management. Proposed mechanism diagram of cyclical elimination of Pb^0 and I^0 defects and regeneration of Eu^{3+} – Eu^{2+} ion pair¹³⁵ (left). Schematic illustration of the Sn^{4+} scavenging method¹⁴⁰ (right). 1,4-Bis(trimethylsilyl)-2,3,5,6-tetramethyl-1,4-dihydropyrazine (TM-DHP) is added to the Sn-based metal halide perovskite (MHP) precursor solution. Sn^0 nanoparticles formed by the reduction of SnF_2 by TM-DHP scavenge residual Sn^{4+} impurities. **d**, Phase conversion. Excess PbI_2 and non-perovskite phase (such as the δ phase) of MHPs can be converted into photoactive 3D MHPs and low-dimensional MHPs by tailored passivators. IPFB, iodopentafluorobenzene. Part c (left) adapted with permission from ref. 135, AAAS and part c (right) adapted with permission from ref. 140, CC BY 4.0.

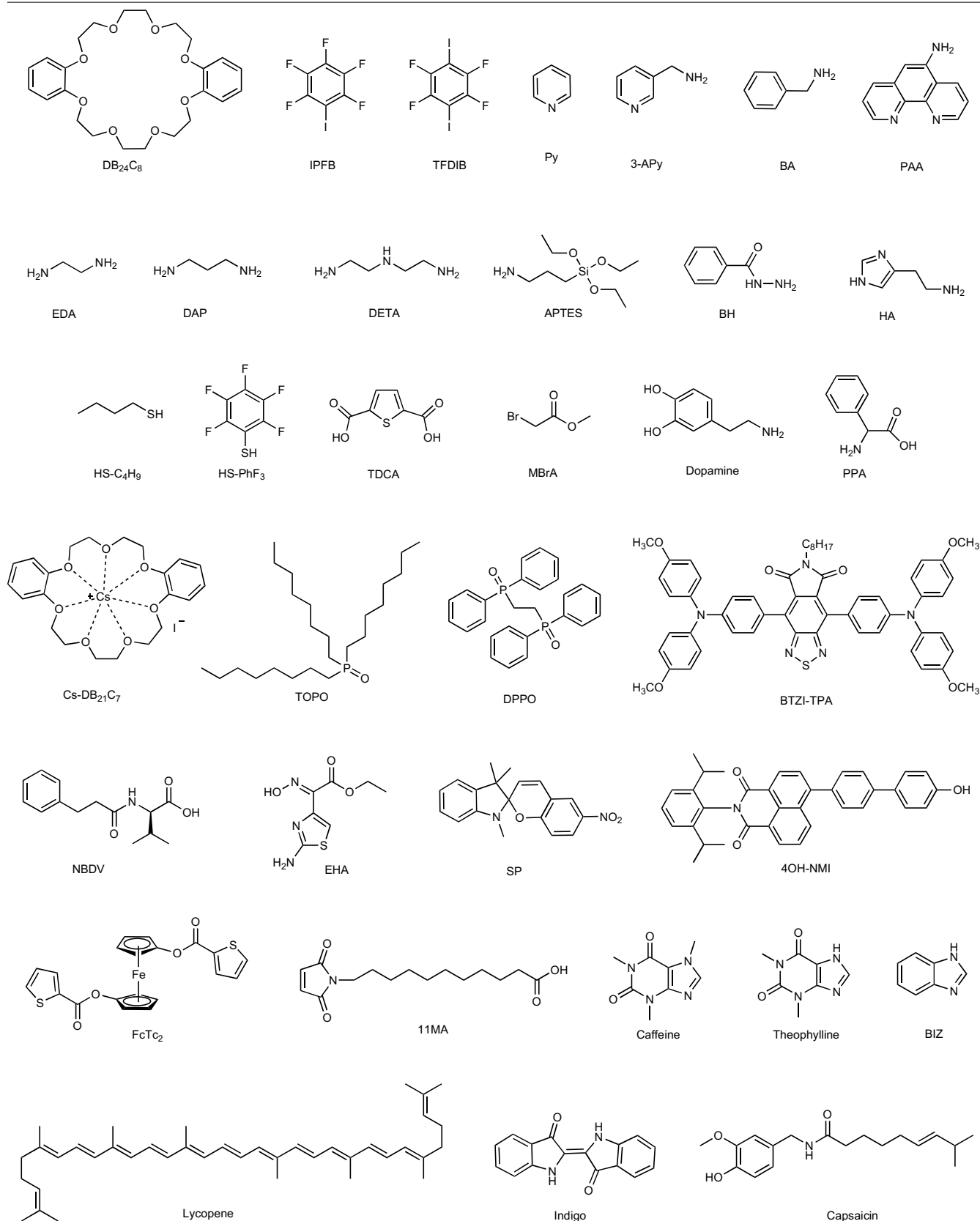


Fig. 4 | Molecular structures of representative examples of organic small molecules used as passivators. Supramolecular agents: IPFB, TFDIB, DB₂₄C₈, Cs-DD₂₁C₇ and TDCA. Lewis bases: Py, 3-APy, BA, PAA, HA, EDA, DAP, DETA,

APTES, TOPO, DPPO, EHA, 11MA, FcTc₂, 4OH-NMI, BIZ, BH, HS-C₆H₉ and HS-C₆F₅. Reducing agent: SP. Natural molecules: caffeine, theophylline, lycopene, indigo and capsaicin.

Phase conversion

Nanoscale active-layer impurities, including hexagonal MHP polytypes and PbI₂ inclusions, are traps for photoexcited carriers – reducing the performance of PSCs¹⁴³ – and also – through the same trapping process – sites at which photochemical degradation of the absorber layer is seeded. Taking excess PbI₂ impurities as an example, on the one hand, a tiny amount of excess PbI₂ suppresses non-radiative recombination and thus improves the relevant device performance¹⁴⁴. On the other hand, photo-sensitive PbI₂ can also decompose to metallic lead (Pb⁰) under illumination, which accelerates the degradation of MHP films and the corresponding devices¹³⁴. To solve this issue, a widely used strategy is to convert excess PbI₂ into different inactive materials, such as 2D, 1D and 0D – meaning layered, wide-bandgap MHPs^{145–149} or other non-MHP materials^{150,151} (Fig. 3d). Non-photoactive MHP impurity phases (6H, 4H and 2H) can also be effectively converted into a black photoactive MHP phase via Cs post-deposition treatment¹²⁸. Incorporation of additional halides (such as 1,8-diiodooctane) into the perovskite precursor solution is another effective strategy for reducing the density of Pb⁰ in the prepared MHP films via ion exchange processes¹⁵².

Design of effective passivators

On the basis of the different passivation mechanisms, organic and inorganic passivators with different molecular structures and compositions have been intensively investigated to improve the efficiency and/or stability of PSCs. The passivators can be classified into four categories: small organic molecules, organic ammonium salts, polymers and inorganic salts.

Small organic molecules

It is hypothesized that an excess of uncoordinated ions (for example, halide anions and Pb cations) is responsible for forming trap states on the surface of the MHP. To passivate the undercoordinated halide anions, supramolecular agents (such as IPFB¹²¹ or 1,2,4,5-tetrafluoro-3,6-diiodobenzene¹²² and crown ethers¹²³) have been proposed to coordinate with these iodine-trapping sites and greatly increase PSC performances. For uncoordinated metal cations (which can be regarded as Lewis acids), a series of Lewis bases acting as passivators have been used to treat MHP films to passivate these surface defects^{153–156} (Fig. 4). For example, Snaith and co-workers demonstrated that pyridine is effective at passivating trap states on the surface of MHP films owing to its interaction with undercoordinated lead atoms¹⁵⁷. Wang et al. reported that benzylamine successfully passivates surface trap states, reducing the number of recombination centres and improving device performances through the interaction between the amino-anchoring group and the Pb-I lattice¹⁵⁸. Combining the advantages of pyridine and amino groups, Jiang et al. reported a simple post-growth treatment of MHP films with 3-(aminomethyl)pyridine¹⁵⁹. The 3-(aminomethyl)pyridine molecule selectively interacts with surface FA⁺ to form adducts, which increases the formation energy of charged iodine vacancies, leading to effective *n*-type doping with a reduced work function in the surface region. The resulting *p*–*i*–*n* structured PSCs obtained a PCE of more than 25% (a certified value of 24.6%, active area: 0.058 cm²), along with excellent operational stability. Similarly, 1,10-phenanthroline-5-amine was explored for synchronous passivation of Pb-I antisite (Pb_i) and

iodine-vacancy (V_i) defects¹⁶⁰. An environmentally benign material, histamine (HA), has also been used to passivate V_i in MHP films via a Lewis acid–base interaction¹⁶¹. Applying a monolayer of bilateral amino groups has also been demonstrated to effectively passivate defects at MHP interfaces and enhance MHP stability by exposing the linking hydrophobic alkyl chains^{162–164}.

Small organic molecules containing oxygen electron lone pair donors have been shown to effectively passivate cationic defects and undercoordinated cation sites in MHPs^{160,165–168}. For example, Krishna et al.¹²⁶ reported that 2,5-thiophenedicarboxylic acid effectively enhances the stability of FAPbI₃-based PSCs by simultaneously passivating various cationic and anionic defects via coordination and hydrogen bonding. 2,5-Thiophenedicarboxylic acid was also used to passivate surface V_i defects in inorganic MHPs and achieve high-performance CsPbI_{2.25}Br_{0.75} PSCs with a remarkable V_{OC} of 1.43 V, reaching 92% of the Shockley–Queisser V_{OC} limit¹⁶⁹.

Hydrophobic thiols (such as pentafluorobenzenethiol) were initially reported as enhancing the moisture stability of PSCs via Pb-S coordination¹⁷⁰. Later, by anchoring a self-limiting passivator (1-butanethiol) to the surface of a 1.68-eV wide-bandgap MHP, Hou et al.¹⁷¹ demonstrated an enhanced diffusion length and suppressed phase segregation, enabling an independently certified PCE of 25.7% for a perovskite-silicon tandem solar cell. Trioctylphosphine oxide featuring a P=O bond has been developed to selectively coordinate through the O atom to the halide-deficient sites of MHP surfaces¹⁷². A diphenylphosphine oxide compound, ethane-1,2-diylbis(diphenylphosphine oxide), coordinates to lead iodide and forms a cross-linked 1D Pb₃I₆–ethane-1,2-diylbis(diphenylphosphine oxide) complex¹⁷³. These judiciously designed cross-linked non-ionic low-dimensional (LD) lead halide or organic adducts can passivate the defects of MHPs while acting as robust ion diffusion barriers, thus greatly improving the electronic quality and intrinsic stability of MHP films.

Given that multiple types of defect coexist in MHPs, functionalizing passivators with multiple interaction sites is an effective strategy to suppress the various defects¹⁷⁴. Yu et al. reported an oxime acid-based material, ethyl 2-(2-aminothiazole-4-yl)-2-hydroxyiminoacetate, with different coordination sites for stabilizing the PbI₂ colloidal clusters, resulting in a very low GB density when growing MHP films, and effectively passivating both Pb-I antisite and iodine vacancy defects¹⁷⁵. In a different approach, Zhang et al. proposed a facile and universal post-treatment strategy for Sn-based perovskites comprising sequential passivation with acetylacetone and ethylenediamine¹⁷⁶. They show that acetylacetone can reduce the trap density and enlarge the grain size, whereas ethylenediamine can bond to undercoordinated Sn and regulate the energy level. A promising 13% efficiency is achieved with better stability in these lead-free PSCs.

Learning from plants, a series of natural molecules, such as indigo¹⁷⁷, lycopene¹⁷⁸, capsaicin¹⁷⁹, caffeine¹⁸⁰ and theophylline¹⁸¹, have also been used as effective passivators for achieving high-performance and operationally stable PSCs. An in-depth study by Wang et al.¹⁸¹ investigated the effects of varying the number and position of Me substituents in the series theophylline, caffeine and theobromine. When a free N–H and C=O were in an optimal relative position in the molecule, hydrogen-bond formation between N–H and I assisted the primary C=O

bonding with an antisite Pb defect to maximize surface-defect binding. Bao and co-workers observed the complete transformation of MAPbI₃ surface energetics from *p*-type to *n*-type during defect passivation with capsaicin, attributed to the spontaneous formation of a *p*–*n* homojunction in the MHP layer¹⁷⁹. This transformation in combination with defect passivation promotes charge transport in the bulk MHP layer and at the MHP–phenyl-C61-butyric acid methyl ester interface, suppressing both defect-assisted recombination and interface carrier recombination. As a result, a 21.88% efficiency and an 83.81% fill factor with excellent device stability were achieved, which were records for polycrystalline MAPbI₃-based *p*–*i*–*n* structured PSCs at the time.

Under operating conditions, dynamic generation of photoinduced defects and free-moving halogen ions were observed in perovskite-active layers. A classical photoisomeric molecule, spiropyran, was used to address this challenge¹⁸². Spiroprans work as light-triggered, self-healing and sustainable passivation sites that continuously repair defects. Li et al. reported a 25% PCE and stable *p*–*i*–*n* structured PSCs by functionalizing the interface with an organometallic compound, ferrocenyl-bis-thiophene-2-carboxylate (FcTc₂), that provided strong chemical Pb–O bonding to reduce surface trap states and accelerated interfacial electron transfer through its electron-rich and electron-delocalizing ferrocene units¹⁸³. The improved interface binding and carrier transport properties of FcTc₂ contribute to superior device stability and a PCE of 25.0% (a certified value of 24.3%, active area: 0.08 cm²) and maintained >98% of the initial efficiency in operational stability tests with a continuous 1-sun illumination for more than 1,500 h.

Organic ammonium salts

Organic ammonium salts have been widely investigated as another type of effective passivators to improve the performance and stability of PSCs^{184–199}. These salts passivate defects through two different mechanisms: (i) by converting excess PbI₂ on the MHP surface, forming a heterogeneous grain structure or a thin layer of a wide-bandgap 2D MHP or transforming a 3D MHP into its 2D counterpart through cation exchange reactions^{11,145,147,200–202} and (ii) by forming a thin organic ammonium salt layer^{120,203}. The general formula of 2D MHPs is (RNH₃)₂(A)_{*n*–1}M_{*n*}X_{3*n*+1}, in which RNH₃⁺ is a primary aliphatic or aromatic alkylammonium cation acting as a spacer between neighbouring MHP layers, A is a small organic cation (for example, FA⁺), M refers to a metal cation such as Pb²⁺ and Sn²⁺, X is a halide or pseudo-halide anion and *n* is the number of [MX₆]^{4–} octahedra within each quantum well³⁷. The hydrophobic nature of spacer cations gives 2D MHPs superior moisture stability⁴⁰. However, the introduction of organic spacers with long alkyl chains decreases the absorption coefficient and carrier transport efficiency of MHP materials and thus undermines the PV performance¹¹. Therefore, researchers have devoted their efforts to maximizing the PV performance of PSCs using 2D MHP passivation layers through molecular design. The molecular structures of organic spacers used in 2D MHPs for PVs are summarized in Fig. 5.

Most MHP surface passivation treatments use either an aliphatic ammonium halide or an aromatic ammonium halide, with butylammonium bromide²⁰⁴ and phenylethylammonium iodide (PEAI)²⁰³ as common representatives. For example, BA⁺ was found to be able to form a thin 2D passivation layer on perovskite surfaces via a solution-based process. Apart from effective surface passivation, the in situ generated, thin 2D layer based on BA⁺ also functions as an electron-blocking layer as well as a humidity barrier, effectively suppressing charge recombination and increasing stability against ambient moisture. The chain

length of organic ammonium salt is a crucial factor concerning defect passivation^{205,206}. For example, *n*-octylammonium iodide delivered an impressive PCE of 25.6% (a certified value of 25.4%, active area: 0.08 cm²) in SnO₂-based FAPbI₃ PSCs by Kim et al.²⁰⁷. Utilizing organic ammonium salts containing long alkyl chains for in situ growth of 2D MHP passivation layers on top of 3D MHP films is not only an efficient strategy for simultaneously passivating both organic cation (such as FA⁺) and halide vacancies but also with regard to improving the moisture stability of MHP owing to the hydrophobic spacer²⁰⁸. The passivation capability of organic ammonium salts is also exceedingly sensitive to their steric properties¹⁴⁵. As a structural isomer of butyl ammonium iodide, bulkier *iso*-butyl ammonium iodide was also explored and proved to be a more effective passivator for improving the PV efficiency and stability of FAPbI₃ PSCs²⁰⁹. Li et al. reported a novel bifunctional molecule, piperazinium iodide, containing both an R₂NH group and an R₂NH₂⁺ group in the same six-membered ring generating an in-plane dipole, and capable of functioning as both an electron donor and an electron acceptor, to react with different surface-terminating groups¹⁴⁸. The PCE of piperazinium iodide-passivated inverted (*p*–*i*–*n*) PSCs reached 23.37% (a certified value of 22.75%, active area: 0.10 cm²), which was the highest value for inverted PSCs at the time. Molecular passivators with large organic substituents normally introduce charge extraction barriers, thus compromising the PV performance of the corresponding PSC devices, although better protection of the MHP is anticipated simultaneously²¹⁰. Apart from the size effect, aliphatic ammonium halides bearing multiple functionalities have also been studied^{120,186,187}, aiming to involve additional functionalities and interactions to improve the PV performance of PSCs. Cyclic aliphatic compounds containing at least one all-carbon ring, which can be either saturated or unsaturated but does not have aromatic character, have also been used as effective passivators in PSCs^{211–215}. Diammonium halide salts have been investigated to passivate MHP surface defects and achieve a more uniform spatial distribution of the surface potential. Chen et al. introduced a thin passivation layer of 1,3-propane diammonium 2D MHP into a monolithic all-perovskite tandem, achieving a record *V*_{OC} of 2.19 V and >27% PCE (a certified value of 26.3%, active area: 0.049 cm²)²¹⁶.

Another variety of organic ammonium salts contains aromatic moieties, among which PEAi is the most studied and used material^{72,202,203,217,218}. Although many reports discuss their hydrophobic properties²⁰³ and how these compounds act as spacers, the electronic effects of aromatic passivators have started to draw attention owing to a potentially more favoured hole extraction from the MHP layer into the neighbouring carrier-transporting layers. It is worth mentioning that the aliphatic spacer between the aromatic ring and ammonium halide is considered crucial because of the π -electron delocalization effect of the ammonium halide. Owing to size limitations, the aromatic substituents used are generally small, such as phenyl^{184,219}, with bandgaps larger than 3.0 eV. As a result, potential electronic effects are not studied, instead research is focused on spacer and hydrophobicity effects. The spacer effect is similar to that of aliphatic ammonium halide passivators; however, the phenyl ring simplifies chemical modification. For instance, Min et al. reported a 25.8% efficient (a certified value of 25.5%, active area: 0.095 cm²) PSC passivated by methoxy (–OCH₃)–substituted PEAi (MeO-PEAi)²²⁰. Yu et al. reported that 4-fluoro-PEAi can be used as a dual-functional passivator for oriented crystallization and comprehensive passivation of wide-bandgap MHPs. Consequently, the modified 1.75-eV WBG PSCs exhibited an improved *V*_{OC} of 1.3 V, leading to a remarkable PCE of 19.1%

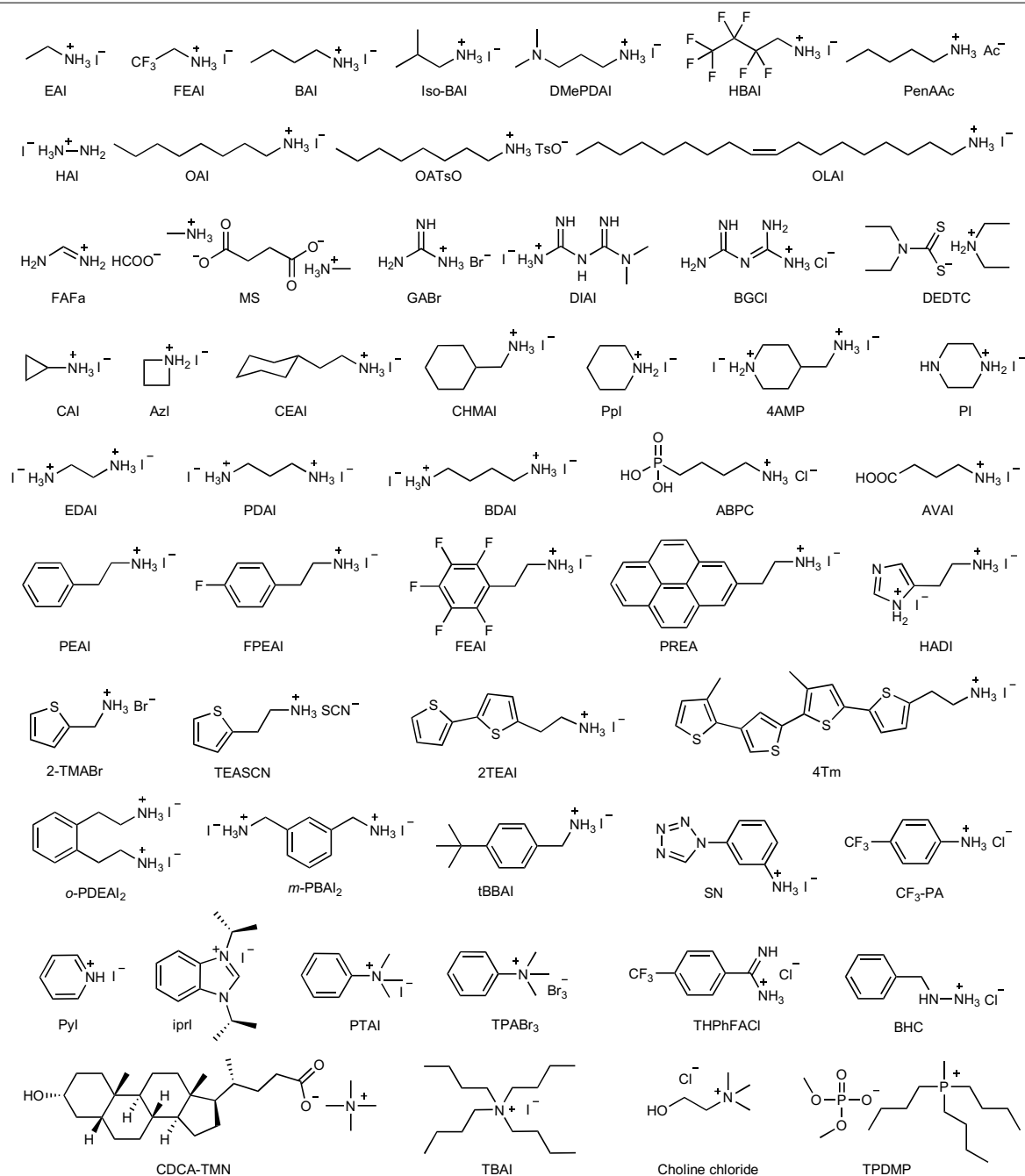


Fig. 5 | Molecular structures of representative examples of organic ammonium salts used as passivators. Formation of a thin organic ammonium salt layer: FAFa, MS, DIAI, BGCI, DEDTC, CF₃-PA, iprl, PTAI, TPABr₃, *o*-PDEAI₂,

THPhFACl, BHC, SN, CDCA-TMN, TBAI, TPDMF, choline chloride. Formation of low-dimensional wide-bandgap perovskite interlayers: the left molecules presented in Fig. 5.

with negligible hysteresis¹⁸⁴. Tailored hydrophobic organic ammonium salts, such as phenyltrimethylammonium iodide, have also been demonstrated to form a stable LD MHP passivation layer, which improves the phase stability of CsPbI₃ and effectively suppresses non-radiative recombination¹⁸⁵. These LD MHP-modified inorganic PSCs based on CsPbI₃ exhibited a record efficiency of 21.0% with high stability.

Investigations into the structural isomers of phenylethylammonium and phenylammonium salts forming LD MHPs¹⁹⁰ or ammonium salts thin layers^{221,222} for effective passivation of MHP defects at interfaces have also been conducted. Incomplete adsorption of passivators into the defective sites on MHPs during passivation treatment is another important factor limiting the performance enhancement of PSCs.

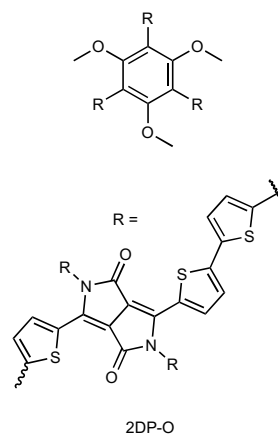
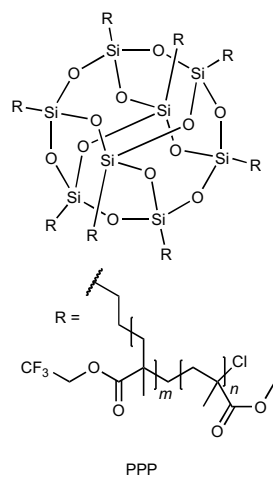
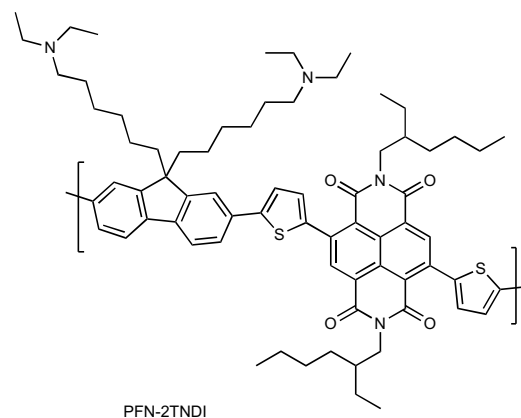
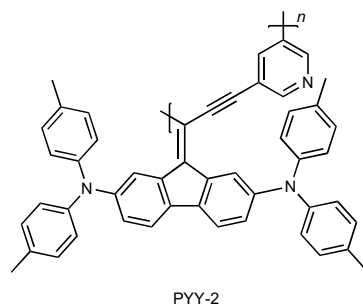
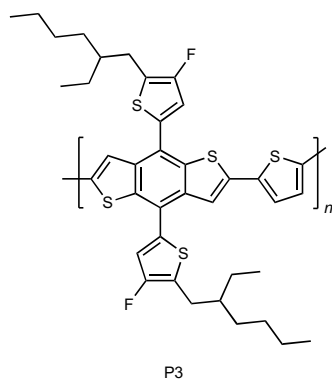
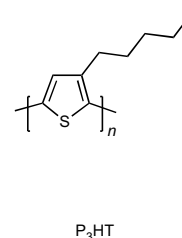
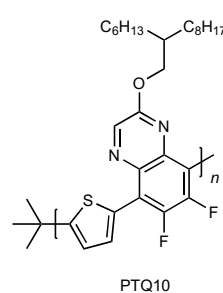
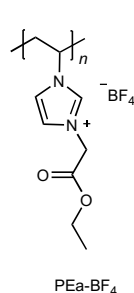
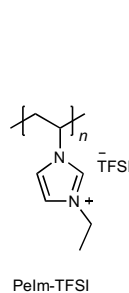
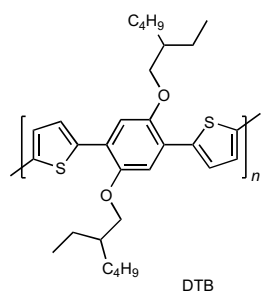
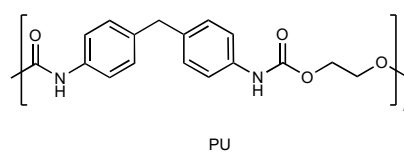
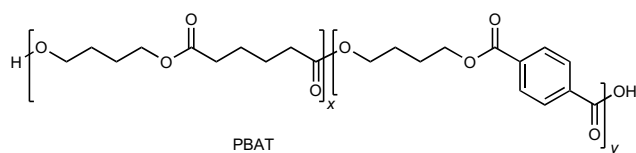
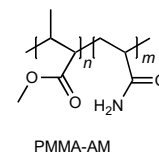
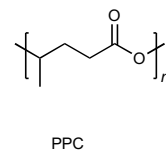
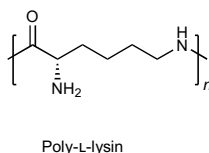
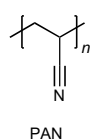
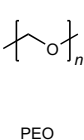
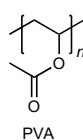
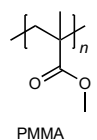


Fig. 6 | Molecular structures of representative polymeric passivators.

Linear polymers: PMMA, PVA, PEO, PAN, PPC, PLL, PMMA-AM, PBAT, PU, and DTB. Two-dimensional polymer: 2DP-O. Three-dimensional polymer: PPP.

Poly(ionic liquid): Pelm-TFSI and PEa-BF₄. Functionalized carrier transport materials: PTQ10, P₃HT, P3, PYY-2 and PFN-2TNDI.

Tan and co-workers reported a strategy to enhance the adsorption and coverage ratio of passivators on Pb-Sn narrow-bandgap MHP films with 4-trifluoromethyl-phenylammonium chloride²¹⁹. By adding a small amount of 4-trifluoromethyl-phenylammonium chloride into the MHP precursor solution, the carrier diffusion length in Pb-Sn MHPs was increased twofold, to more than 5 μm , which contributed to a certified efficiency of 26.4% (active area: 0.049 cm^2) in all-perovskite tandem solar cells, exceeding that of the best-performing single-junction PSCs. In addition to phenyl-containing aromatic passivators, other aromatic ammonium salts containing thiophene^{189,192,223}, pyrimidine¹⁹⁷, imidazole^{196,224} and pyrene²²⁵ have also been investigated as effective passivators.

In addition to the cation backbone of alkylammonium salts, counterions also have considerable effects on device performance and longevity. In most cases, iodide (I^-), bromide (Br^-) or chloride (Cl^-) are used. I^- ions have demonstrated a promising capacity for the passivation of halide vacancies owing to their compatibility with iodide-rich MHPs. Up to now, the majority of high-performing passivators use iodide as counterion²²⁰. Owing to a reduced orbital overlap between the halide and Pb^{2+} , Br^- -based perovskites normally exhibit a larger bandgap compared with I^- . As a result, ion exchange between the halides in the MHPs and the halides in the ammonium salts causes band bending at the interfaces, which can increase the V_{oc} of such PSCs²²³. Cl^- ions in ammonium salts can exchange with halides in the MHP network and slow down crystallization rates to obtain larger grains and better morphologies¹⁹⁷. Polyatomic anions have also been investigated to effectively passivate halide vacancies, thereby stabilizing high-performance PSCs^{117,149,188,189,195,226–229}. A study showed that substituting I^- with tosylate (TsO^-) fully neutralizes the negative work function change caused by surface passivation treatment, which lowers the halide migration activation energy, and greatly improves PSC stability²²⁶. Through synthetic manipulation of anions and cations, Gao et al. reported that pentylammonium (PenA^+) and acetate (Ac^-) exhibit strong chemical bonding to both negative and positive defects of MHP films. PenAAc-modified flexible PSCs achieved a record PCE of 23.68% (a certified value of 23.35%, active area: 0.08 cm^2) with improved operational stability²²⁷. An ammonium–pseudo-halide ion pair compound has been developed for providing a synergistic chemical passivation for either cationic or anionic defects in FAPbI_3 (ref. 198) and $\text{CsPbI}_{3-x}\text{Br}_x$ MHP films¹⁹⁹. Mixing two ammonium salts is another strategy to passivate multiple types of defect in MHPs^{193,194,230}.

The protic polar solvents commonly used for dissolving the organic ammonium salts, mainly isopropyl alcohol, result in a destabilization of the underlying MHP layer and undesirable degradation of devices^{200,231}. Nonpolar solvents, such as chloroform showing very limited solubility towards FAI (<1 mg mL^{-1}) and PbI_2 , were used to selectively dissolve organic ammonium salts and maximize device performance by suppressing the formation of the MHP δ -phase and amorphous phase during surface passivation treatment²⁰⁰. Sidhik et al.²³² found that solvents with an appropriate dielectric constant and donor strength could grow phase-pure 2D MHPs with controlled thickness and composition on as-prepared 3D MHPs without damaging or dissolving them. The fabricated PSCs maintained their initial

PCE of 24.5% for 2,000 h with <1% efficiency loss under continuous operation at 55 $^\circ\text{C}$ and 65% relative humidity. To avoid solvent effects, Choi et al. reported a vacuum deposition strategy for growing a highly oriented 2D MHP as a surface passivation layer via bottom-up growth on the bulk perovskite absorber layer²³³. By leveraging the passivating properties and controlling crystallinity, vacuum-deposited 2D MHP considerably reduces the trap density of bulk perovskite and enhances charge transport, resulting in a 21.4% PCE of vacuum-processed PSCs with enhanced operational stability.

Polymers

Compared with other passivators, polymers have unique advantages when used for defect passivation in PSCs. Polymeric passivators are beneficial for durable defect passivation under operational conditions owing to their bulky and long-range-ordered molecular structure as well as multiple chemical-anchoring sites. Polymers containing functional groups (such as carbonyl^{234–238}, amine^{239,240}, pyridine^{241,242}, thiophene^{243,244} and cyano²⁴⁵) are effective passivators, which interact strongly with MHPs via chemical coordination bonding, successfully decrease surface defects and suppress non-radiative recombination losses. To improve the interaction between polymer and MHP, the design of novel polymers with multiple functionalities for surface passivation has attracted increasing attention combining high efficiency and long-term stability of PSCs (Fig. 6). Zhang et al. reported a multi-functional poly(methyl methacrylate-co-acrylamide) polymer that effectively passivates interfacial defects²⁴⁰. The coexisting ester and amide groups coordinate to Pb^{2+} at the interface and also anchor the I^- to suppress defects such as interstitial I^- and Pb-I antisite. It is worth emphasizing that the formation of N–H...I hydrogen bonds could further assist the primary C=O bonding with Pb^{2+} . This treatment resulted in the PCE of a perovskite module (14 cm^2) reaching 20.64%, accompanied with superior long-term stability. Yang and co-workers developed an in situ polymerization-assisted grain growth strategy to manipulate the crystallization of MHPs and suppress defect formation by incorporating monomers into precursor solutions^{235,246}. Poly(ionic liquid) is another emerging group of polymer passivators that efficiently promote interfacial charge extraction and reduce the density of trap states^{247,248}. Strategic introduction of passivating functional groups into polymeric carrier transport layers has also been widely investigated, which simultaneously minimize interfacial non-radiative recombination and promote charge extraction and transport^{242,243,249–252}. For example, Zhu and co-workers reported a dopant-free pyridine-based polymer hole transport material (PPY2), which contains energy levels compatible with that of MHPs, high hole mobility, effective passivation of undercoordinated Pb^{2+} and iodide defects, as well as the capability to promote the formation of high-quality MHP films. The PPY2-based inverted PSCs delivered an encouraging PCE of up to 22.41% with a small V_{oc} loss (0.40 V), among the best device performances for inverted PSCs reported at the time. In addition to the linear polymers, star-shaped²³⁷ and 2D²⁵³ polymers can also be used as effective passivators. Cao et al. developed an ‘all-in-one’ passivation strategy realized by synthesizing a star-shaped polyhedral oligomeric silsesquioxane-poly(trifluoroethyl methacrylate)-b-poly(methyl methacrylate) (PPP) polymer with

multiple functional groups – including C=O, $-\text{CF}_3$ and $\text{C}-\text{O}^{237}$. This PPP polymer with multiple chemical anchoring sites in the star-shaped branches strongly manipulates the crystallization of MHP films, resulting in a lower trap density and higher carrier mobility by inhibiting non-radiative recombination that reduces charge-transport losses. PPP polymer-modified inverted PSCs showed a PCE of 22.1%, and a very high fill factor of 86.2% as well as an excellent long-term operational stability (no efficiency loss after 1,000 h of continuous operation at 45 °C under one-sun illumination). In summary, rationally designing polymeric passivators with suitable functional groups is eminently effective in improving device performance and longevity²³⁸.

Inorganic salts

Although many organic molecules can passivate MHP defects through hydrogen bonds, coordination bonds or ionic bonds, these bonds can still lead to stability issues related to ambient and thermal stress. Inspired by the stable nature of inorganic materials with strong chemical bonds and high stability, Yang et al. first reported a general passivation strategy for MHPs by forming a thin, compact wide-bandgap inorganic lead oxysalt layer (such as PbSO_4) on an MHP surface through in situ reactions with selected inorganic anions¹⁴⁶. The surface lead oxysalt layer formed strong bonds with the MHP and afforded much better resistance to many detrimental stimuli under ambient atmosphere and light illumination. These layers also reduced the defect density on MHP surfaces by passivating undercoordinated surface lead centres, thereby improving device performance and operational stability. Li et al. reported a surface sulfidation treatment to form a stable PbS passivation layer on MHP films through an in situ reaction between lead(II) di-(pyridine-2-carboxylate) ($\text{Pb}(\text{PyA})_2$) and hexamethyldisilathiane²⁵⁴. The strong lead–sulfur ($\text{Pb}-\text{S}$) bonds that are formed elevated the Fermi level at the MHP interface and induced an extra subsurface field for electron extraction, which contributed to the fabrication of highly efficient $p-i-n$ structured PSCs with PCEs >24% (a certified value of 23.47%, active area: 0.08 cm^2). Wang et al. developed a chlorinated graphene oxide passivation layer to stabilize the soft MHP heterojunctions with strong $\text{Pb}-\text{Cl}$ and $\text{Pb}-\text{O}$ bonds²⁵⁵. The obtained heterostructure can selectively extract photogenerated charge carriers and impede the loss of decomposed components from soft MHPs, thereby reducing damage to the organic charge-transporting semiconductors. Another effective inorganic passivator is KI, capable of passivating undercoordinated halides at the surface and GBs via ionic interactions as reported by Abdi-Jalebi et al.²⁵⁶. Through bonding with halides, K^+ is also effective in suppressing ion migration and phase segregation of mixed-halide MHPs. Other metal ions, such as Cu^+ , Cd^{2+} and Nd^{3+} , are capable of effectively passivating defects in the bulk or at the surface and GBs^{76,257–259}. As we discussed before, the secondary-phase excess PbI_2 in MHP films has some positive effects on PV performance but can be detrimental to device stability. You et al. reported that rubidium chloride (RbCl) can be used to convert excess PbI_2 into an inactive $(\text{PbI}_2)_2\text{RbCl}$ compound, which effectively stabilizes the MHP phase and delivered a PCE of 26.1% (a certified value of 25.6%, active area: 0.074 cm^2) for $n-i-p$ structured FAPbI_3 solar cells¹⁵⁰.

Conclusions and outlook

Passivation of MHPs has already improved substantially with regard to initial efficiency and long-term stability of PSCs. Nonetheless, we are convinced that there is still considerable room for further development. In this Review, we discussed the main types of defect in MHPs, their detrimental impact on PSC performance as well as passivation mechanisms

used to mitigate these effects. The most effective passivators reported to date were presented, categorized into four classes – small organic molecules, organic ammonium salts, polymers and inorganic salts. By reviewing the structures and properties of these compounds, we provide a basis for the rational design of optimized passivators for future applications. In addition, we point out four avenues for future research that will be crucial pursuits for the field to leverage the concept of passivation and overcome the limitations still faced by PSCs.

First, it will be crucial to get a clearer picture of the atomistic structure of interfaces and GBs. To date, most studies in this area have focused on a purely computational approach, although there is still a lack of hard experimental evidence. The nature of the defects present at these sites as well as their densities, spatial distribution and energy levels of the associated trap states need to be known to conceive tailor-made passivation strategies. Related to this, it is also essential that we better understand the mechanisms of passivation, as this will greatly support future efforts to optimize passivators. This includes both the mechanisms leading to improved initial efficiencies and better long-term stabilities.

Second, we suggest it to be crucial to use high-throughput experimental techniques with machine-learning algorithms to find the optimum structures of passivators²⁶⁰. Most publications only report results on a small number or even just one compound, and considerable variations with regard to the experimental details as well as results can be found between different laboratories. It is, therefore, difficult to directly compare results obtained by different groups stifling efforts to compile the large sets of consistent data necessary for computational analysis. Machine learning is a highly promising approach to solve the multiparameter problem of optimizing the structure of passivators, considering their effects on the perovskite material as well as device properties with regard to initial efficiency and long-term stability.

Third, we note that more in-depth studies are especially needed for certain interfaces, for which passivation has so far only shown limited success. One such example is the interface between perovskite and fullerene-derived electron transport layers in inverted-type PSCs. In this regard, the use of multifunctional passivators opens promising new areas for future studies.

Finally, as passivation is likely to have a key role in the envisioned commercial production of PSCs, it is important to develop the scaling-up techniques necessary to integrate this step into industrial production lines. Here it would be preferable if passivation could be achieved in the same step as perovskite formation by including the passivating material directly into the perovskite precursor.

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Author contributions

H.Z. and L.P. contributed equally to this work. M.G. and S.M.Z. contributed to the writing and editing of this manuscript. H.Z. and L.P. researched the literature for the article and contributed to the discussion of content and writing. J.C. contributed to the discussion and reviewing of the manuscript.

Competing interests

The authors declare no competing interests.

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